

FANUC Robot R-J3/R-J3iB Servo Gun : Influence by the friction increase

1. Outline

The following troubles for the above system have been reported. It is caused by decreasing of temperature. Then, especially in the Monday morning, these could occur.

- 1 Pressing motion is slow.
- 2 Error excess alarm (SRVO-023 or SRVO-024) occurs during pressing motion.
- 3 Measurement of tip wear down does not work correctly or does fail.

2. Cause of problem

These problems are caused by the increase of friction according to the temperature decrease. For the servo gun control, the friction is calculated beforehand by the auto-tuning program. However, this friction has changed more than amount of the assumed change (+30%) from executing of auto tuning. This change of friction is the reason of above problems.

3. Workaround

When these problems occur, please apply the following countermeasures;

Workaround 1:

- (1) Change the servo parameter that specifies the allowable amount of friction.
 Servo Param [n] [183] = About 1.3 times of the former value (Consider the sign)
 n : The hardware axis number for servo gun
 (Please refer to next page about the method of displaying the servo parameter)
- (2) Change the panel detection threshold value for measurement of tip wear down.
 \$SGGUN1.\$SGREG.\$REGLONG[1] = former value + about 15(kgf)
 (\$SGGUN2... in case of gun 2)

Workaround 2:

Please re-tune the gun when the problem still occurs frequently after workaround 1.

- (1) Re-execute the servo gun auto-tuning program and re-calibrate the welding pressure.
The auto-tuning program changes the welding pressure calibration to "incomplete".
Please re-calibrate the pressure if necessary.
Moreover, the gun sag compensation might become "disabled" in some versions of the software. Please confirm this setting after completing the auto-tuning.
- (2) Reset the value of tip wear down by using new tip.
Please refer to "Chapter 5.3 RESET" of "Servo Gun Function Supplementary Manual"(TMR02065E B-81514EN@01-3) for details.

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Edit	Date	Design	Description	FANUC LTD.		PAGE	1/2	

Appendix:

Servo parameter display and modification method

- First, display the system variable screen.
"MENU" key -> "0" (--NEXT--) -> "6" (SYSTEM) -> "F1" ([TYPE]) -> "2" (Variables)
- Change the following system variables.
\$SVPRM_ENB = 1 (Default value = 0)
- Display the servo parameter screen.
"F1" ([TYPE]) -> "3" (Servo Param)
The following screens will be displayed.

PRG001	Line 0	ABORT
System Servo Parameter	JOINT	10 %
11/300		
Servo Param [7] [3] =	00000000	00000000
Servo Param [7] [4] =	00000000	00000000
Servo Param [7] [5] =	00000000	00000000
Servo Param [7] [6] =	00000000	00000000
Servo Param [7] [7] =	00000000	00000000
Servo Param [7] [8] =	00000000	00000000
Servo Param [7] [9] =	00000000	00000000
Servo Param [7] [10] =	00000000	00000000
Servo Param [7] [11] =		
Servo Param [7] [12] =		

Parameter Number

0

- Switch the axis number.
"F2" (AXIS#) -> input the axis number -> "ENTER" key
The axis number is dependent on the system configuration.
Generally, this number is 7 or later for servo gun axis.
- Change the servo parameter
Please move the cursor to the parameter number which needs the change.
Input the new value, and then push "ENTER".
The servo power is turned off automatically when the servo parameter is changed.
There is no necessity to cycle power after parameter change.
- Please close the servo parameter screen by pushing "F5"(DONE) after parameter change.

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